

WASHINGTON RONALD REAGAN, VA, USA

WMO: 724050

Lat: 38.847N

Lon: 77.035W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.3	(c) -6.3	(d) -19.5	(e) 0.7	(f) -6.7	(g) -16.9	(h) 0.9	(i) -4.2	(j) 11.9	(k) 1.1	(l) 11.0	(m) 0.3	(n) 5.3	(o) 330	(p) 0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.7	(c) 34.7	(d) 24.2	(e) 33.3	(f) 23.7	(g) 32.0	(h) 23.1	(i) 25.8	(j) 31.7	(k) 25.2	(l) 30.7	(m) 24.6	(n) 29.6	(o) 4.5	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.2	28.5	23.6	18.4	27.9	23.0	17.7	27.4	79.4	32.0	76.6	30.8	74.1	29.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.5	(b) 9.0	(c) 8.1	(e) -11.0	(f) 36.7	(g) 3.0	(h) 1.8	(i) -13.2	(j) 38.0	(k) -14.9	(l) 39.1	(m) -16.6	(n) 40.1	(o) -18.8	(p) 41.4		
(4) 10.5	(5) 9.0	(6) 8.1	(7) -12.5	(8) 27.2	(9) 2.7	(10) 0.9	(11) -14.5	(12) 27.9	(13) -16.1	(14) 28.5	(15) -17.6	(16) 29.0	(17) -19.6	(18) 29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	2.6	4.2	8.4	14.4	19.3	24.3	26.8	26.0	22.3	15.9	9.7	5.2	(6)
(7)		DBStd	9.45	5.46	5.11	5.07	4.54	4.13	3.26	2.65	2.53	3.49	4.31	4.41	4.59	(7)
(8)		HDD10.0	737	237	173	92	12	0	0	0	0	0	4	58	161	(8)
(9)		HDD18.3	2142	487	397	309	134	39	2	0	0	7	101	259	407	(9)
(10)		CDD10.0	2558	8	10	43	144	289	429	520	497	369	186	50	12	(10)
(11)		CDD18.3	922	0	0	2	16	70	182	262	239	127	24	1	0	(11)
(12)		CDH23.3	8263	0	1	16	137	526	1607	2723	2231	882	137	3	0	(12)
(13)		CDH26.7	3131	0	0	2	33	157	599	1176	864	272	28	0	0	(13)
(14)	Wind	WSAvg	3.9	4.2	4.2	4.6	4.3	3.9	3.7	3.5	3.4	3.6	3.6	3.8	3.8	(14)
(15)	Precipitation	PrecAvg	1017	71	67	85	74	97	98	101	90	93	82	75	82	(15)
(16)		PrecMax	1684	181	144	215	175	271	356	247	233	314	239	192	166	(16)
(17)		PrecMin	722	10	9	1	1	19	24	17	9	3	0	7	11	(17)
(18)		PrecStd	194	33	35	39	34	51	72	54	52	70	53	45	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.6	25.5	30.0	32.4	35.5	37.1	35.9	34.1	29.8	23.4	20.3	(19)
(20)			MCWB	14.3	14.7	16.2	19.2	22.3	24.3	24.7	24.7	23.0	21.9	16.8	15.8	(20)
(21)		2%	DB	15.9	17.3	21.8	27.2	30.2	33.2	35.1	34.1	31.8	26.7	20.4	16.8	(21)
(22)			MCWB	12.9	11.9	14.2	17.7	21.5	23.4	24.4	24.2	22.3	20.6	15.5	13.8	(22)
(23)		5%	DB	13.1	14.0	19.1	24.5	28.4	31.7	33.6	32.5	29.7	24.5	18.4	14.2	(23)
(24)			MCWB	10.1	8.9	12.8	16.7	20.7	22.9	24.1	23.4	21.6	19.2	14.4	11.4	(24)
(25)		10%	DB	10.1	11.4	16.4	22.1	26.4	30.1	32.1	31.0	27.9	22.5	16.5	11.7	(25)
(26)			MCWB	6.9	7.6	11.0	15.3	19.7	22.3	23.6	22.9	21.1	18.0	12.8	8.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.6	17.6	21.3	23.9	26.0	27.0	26.7	25.4	23.6	18.9	17.3	(27)
(28)			MCDWB	17.7	20.3	23.4	26.5	30.1	32.3	33.2	32.9	30.4	26.6	21.1	19.3	(28)
(29)		2%	WB	13.6	12.9	15.7	19.5	22.8	25.1	26.0	25.6	24.4	21.9	17.0	14.5	(29)
(30)			MCDWB	15.9	16.4	20.2	24.4	28.4	30.5	32.0	31.5	28.6	25.4	19.3	16.4	(30)
(31)		5%	WB	10.4	10.1	13.8	17.9	21.7	24.4	25.3	24.9	23.5	20.6	15.2	11.9	(31)
(32)			MCDWB	12.7	12.9	18.2	23.0	26.9	29.6	31.1	30.1	27.4	23.5	17.7	13.9	(32)
(33)		10%	WB	7.6	7.9	11.7	16.5	20.7	23.7	24.7	24.2	22.6	18.8	13.4	9.1	(33)
(34)			MCDWB	9.8	11.2	15.6	21.3	25.1	28.4	30.0	28.5	26.3	21.7	15.8	11.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	9.4	10.2	9.4	9.0	8.7	8.6	8.7	8.4	7.2	(35)	
(36)			MCDBR	11.5	12.6	13.4	13.5	11.6	10.7	10.4	10.2	10.5	10.7	10.9	10.4	(36)
(37)			MCWBR	8.9	8.5	8.0	6.7	5.4	4.3	3.8	3.8	4.3	5.6	7.1	8.4	(37)
(38)		5% WB	MCDWR	10.8	11.0	12.3	11.8	10.5	9.5	9.1	9.0	8.4	9.0	9.4	9.6	(38)
(39)			MCWBR	9.4	8.6	8.6	6.8	5.2	4.3	3.8	3.8	4.2	5.8	7.4	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.323	0.358	0.404	0.445	0.502	0.524	0.518	0.430	0.363	0.325	0.311	(40)	
(41)		taud	2.488	2.430	2.396	2.285	2.202	2.083	2.041	2.039	2.261	2.453	2.516	2.519	(41)	
(42)		Ebn at Noon	869	902	900	873	841	790	769	761	813	845	843	841	(42)	
(43)		Edh at Noon	79	95	109	129	143	162	167	164	124	92	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	2.93	3.90	4.93	5.56	6.14	6.01	5.36	4.37	3.28	2.38	1.84	(44)	
(45)		RadStd	0.17	0.33	0.28	0.37	0.62	0.47	0.31	0.28	0.51	0.36	0.20	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-99	+167	+107
(47) Regional (74 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+74

Nomenclature: See separate page